Colloid

2021

(a) What is colloid? How does it differ from solution?

(b) What do you mean by micelles? Explain with suitable example (2017) (2015)

(c) Discuss electrical double layer for understanding the electrical properties of colloids. How can you measure zeta potential from it?

(d) Define the terms elastic and nonelastic gels with suitable example.

(e) Write the applications of colloids.

2018, 2020

(a) What are colloids? Present the classification of colloids depending on the nature of the dispersed phase and dispersion medium in a tabular format with example (2018)

(b) Distinguish between lyophobic and lyophilic sols. (2017) (2018) (2015)

(c) Discuss the optical and electrical properties of colloidal solution (2018)

(d) What are emulsions? How does the presence of surfactant in the system stabilize the emulsion. (2018)

2017

(a) Illustrate the relationship between electrophoresis and electroosmosis of colloids their reverse phenomenon (2015)

(b) What are the methods available for the preparation of lyophilic and lyophobic sols? Discuss the electro-dispersion method for the preparation of lyophobic sols.

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(*) 2021-a

True Solution: True solution is the homogeneous solution in which the particle size is (0.1nm to 1 nm) and the solution pass through the filter paper

Suspension: (i) The particle size of the suspension is greater than 200 nm
(ii) The particle are visible even with naked eye
(iii) it cannot pass through the filter paper.

Colloid: The particle size of the colloidal solution in the rang of 1nm to 200n

→ They are intermediate between true solutions and suspensions.
→ it can pass through the filter paper
ex→ Fog, Ice-cream, Butter, Soap bubles.
→ Suspension > Colloidal Solution > true solution

⇒ Colloids are unlike solutions because their dispersed particles are much larger than those of the solution that's why colloid differ from solution

(*) 2020-a → The substance distributed as the colloidal particles is called dispensed phase.
→ The second continuous phase in which the colloidal particles are dispersed is called the dispersion medium.
→ There are eight types of colloidal system possible

Dispersed phase	Dispersion medium	name	Examples
Solid	gas	Aerosol	smokes, dusts
Solid	liquid	Sol	Paint, ink
Solid	Solid	— —	ruby glass
Liquid	gas	Aerosol	mists, clouds
Liquid	liquid	Emulsion	milk
Liquid	Solid	Gel	butter
Gas	Gas	— —	Unknown
Gas	liquid	Foam	[illegible]

(*) 2018-b

Sols: Sols are colloidal system in which a solid is dispersed in a liquid

There are two types of sols (i) Lyophilic sols (ii) Lyophobic sols

Lyophilic sols	Lyophobic sols
(i) Lyophilic sols are those in which the dispersed phase attraction for the dispersion medium	(i) Lyophobic sols are those in which the dispersed phase has no attraction for the dispersion medium
(ii) prepared by direct mixing with dispersion medium	(ii) not prepared by direct mixing with the medium
(iii) no charge on particle	(iii) particles carry positive or negative charge
(iv) particles generally solvated	(iv) No solvation of particles
(v) viscosity higher than dispersion medium	(v) viscosity almost the same as of medium
(vi) precipitated by high conc. of electrolytes	(vi) precipitated by low conc of electrolytes
(vii) Reversible	(vi) irreversible
(viii) Do not ~~exhibit~~ show Tyndall effect	(vii) ~~exhibit~~ show Tyndall effect
(ix) particles migrate to anode or cathode, or not at all	(viii) particles migrate to either anode or cathode

(*) 2018-d

Emulsions: emulsions are liquid–liquid colloidal system.

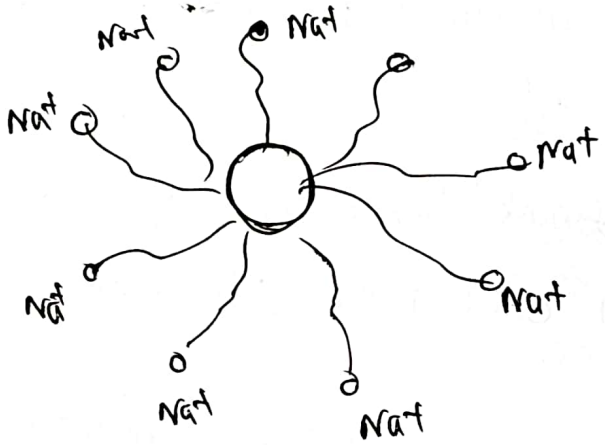
→ emulsion may be defined as a dispersion of finely divided liquid droplets in another liquid.

→ There are two types of emulsions (i) oil in water type → milk
(ii) water in oil type → stiff greases

→ To have a stable emulsion,
Small amount of a third substance called the **emulsifier** is added during the preparation.

→ The emulsifier concentrates at the interface and reduces surface tension on the side of one liquid which rolls into droplets. It is made of a long hydrocarbon tail (oil soluble) with a polar head - $COONa^+$ (water soluble).

→ In o/w type emulsion the tail is attached into the oil droplet.



(*) 2021-d

Gel → A gel is a jelly-like colloidal system in which a liquid is dispersed in a solid medium ex: butter

There are two types of gels (i) Elastic gels (ii) Non-elastic gel

Elastic gels:

→ Elastic gels are those which posses the property of elasticity

→ They change their shape on applying force and return to orginal shape when the force is remove

→ Gelatin, Starch and Soaps are examples of substance which form elastic gels.

→ Elastic gels are obtained by cooling fairly concentrated lyophillic sols.

→ The linkage between the molecules are due to electrical attractio

Non-elastic gels:

→ Non-elastic gels are those which are rigid.

→ Silica gel

→ These are prepared by appropriate chemical action

→ Thus silica gel is produced by adding concentrated hydrochloric acid to sodium silicate solution of the correct concentration.

→ The resulting molecules of silicic acid polymerise to form silica gel

→ it has a network linked by covalent bonds which give a strong and rigid structure.

⊛ 2021-b

⇒ In the solvent the soap or detergent molecules formed colloidal sum is called micelles

→ Sodium stearate

→ $C_{17}H_{35}COO^-Na^+$

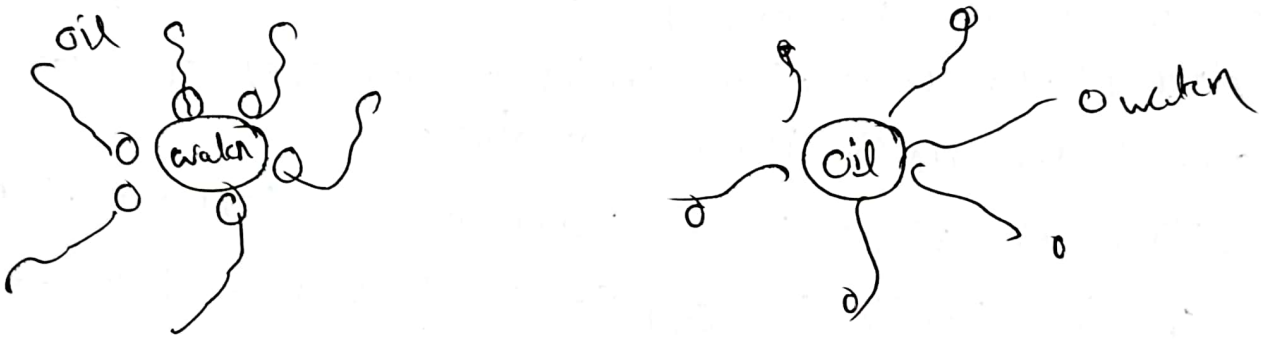
→ The colloidal size of 70 stearate ions group form a micelle

→ The stearate ion has a long hydrocarbon chain (hydrophobic part) with a polar $-COO^-$ group (hydrophilic part) at one end.

→ The zig zag hydrophobic tail is shown by a wavy line and hydrophilic part by a hollow circle

→ in the micelle formation, the tails being insoluble in water are directed toward the center. while the soluble polar heads are on the surface in contact with water.

→ The charge on the micelle due to the polar heads accounts for the stability of the particle



⊛ Application of colloids

Foods:

(i) milk is an emulsion of butterfat in water protected by water

(ii) Casein, Gelatin deserts are example of colloids

(iii) ice cream is a dispersion of ice in cream

medicines:

(i) Halibut-liver oil and cod-liver that we take are, in fact, the emulsions of the respective oils in water

(ii) many ointments for application to skin consist of physiologically active components dissolved in oil and made into an emulsion with water

(iii) Penicillin is produced in colloidal form suitable for injection,

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⊛ Non-drip paints:

→ All paints are colloidal dispersions of solid pigments in a liquid medium

→ The modern non-drip paints also contain long chain polymers. At rest, the chains of molecules are coiled and entrap much dispersion medium.

→ Thus the paint is a semisolid gel structure. When shearing stress is applied with a pain brush. the coiled molecules straighten and the entrapped medium is released.

⊛ Discuss the optical and electrical properties of colloidal solution.

optical properties

Tyndall effect

→ when a strong beam of light is passed through a sol and viewed at right angles. The path of light shows up as a cone. This is due to the fact that sol particles absorb light energy and then emit it in all direction in space. This is called **Scattering of light**. The phenomenon of the scattering of light by the sol particles is called Tyndall effect

→ The illuminated beam or cone formed by the scattering of light by the sol particles is often referred as Tyndall beam or Tyndall cone.

→ if the sol particle are large the sol may even appear turbid in ordinary light as a result of Tyndall scattering

→ True solutions do not show Tyndall effect because is particle size is small

→ Thus Tyndall effect can be used to distinguish a colloidal solution from a true solution

Electrical properties of sol

The sol particles carry an electric charge:

→ The most important property of colloidal dispersions is that all the suspended particles posses either a positive or a negative charge. The mutual forces of repulsion between similarly charge particle prevent them from establishement under the action of gravity. This gives stability to the sol.

→ The sol particle acquire positive or negative charge by adsorption of positive or negative ions from the dispersion medium.

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Electrical Double layer:

→ The surface of colloidal particle acquires a positive charge by selective adsorption of a layer of positive ions around it. This layer attract counterions from the medium which form a second layer of negative charges.

→ The combination of the two layer of +ve and -ve charge around the sol particle was called **Helmholtz Double layer**

→ The double layer is made of

ⓐ a **compact layer** of positive and negative charges which are fixed firmly on the particle surface.

ⓑ a **Diffuse layer** of counterions diffused into the medium containing positive ions.

→ The combination of the compact and diffuse layer is called **Stern Double layer**

→ The diffuse layer is only loosely attached to the particle surface and moves in the opposite direction under an applied electric field.

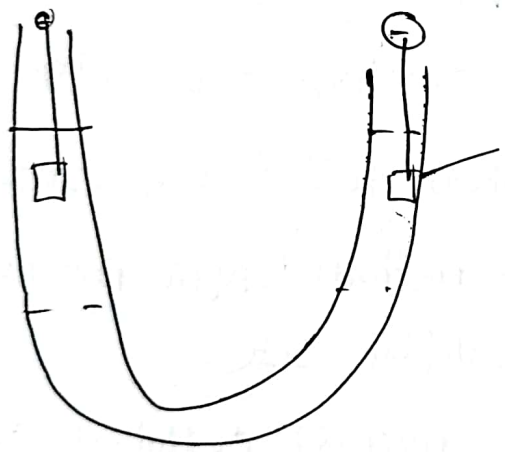
→ Because of the distribution of the charge around the particle, there is a difference in potential between the compact layer and Diffuse layer. This is called **Zeta potential.**

→ it has been made possible to estimate the magnitude of the zeta potential with the help of electrophoresis and electro osmosis properties.

Electrophoresis:

→ if electric potential is applied across two platinum electrodes dipping in a hydrophilic sol, the dispersed particles move towards one or the other electrode.

The movement of sol particles under an applied electric potential is called electrophoresis

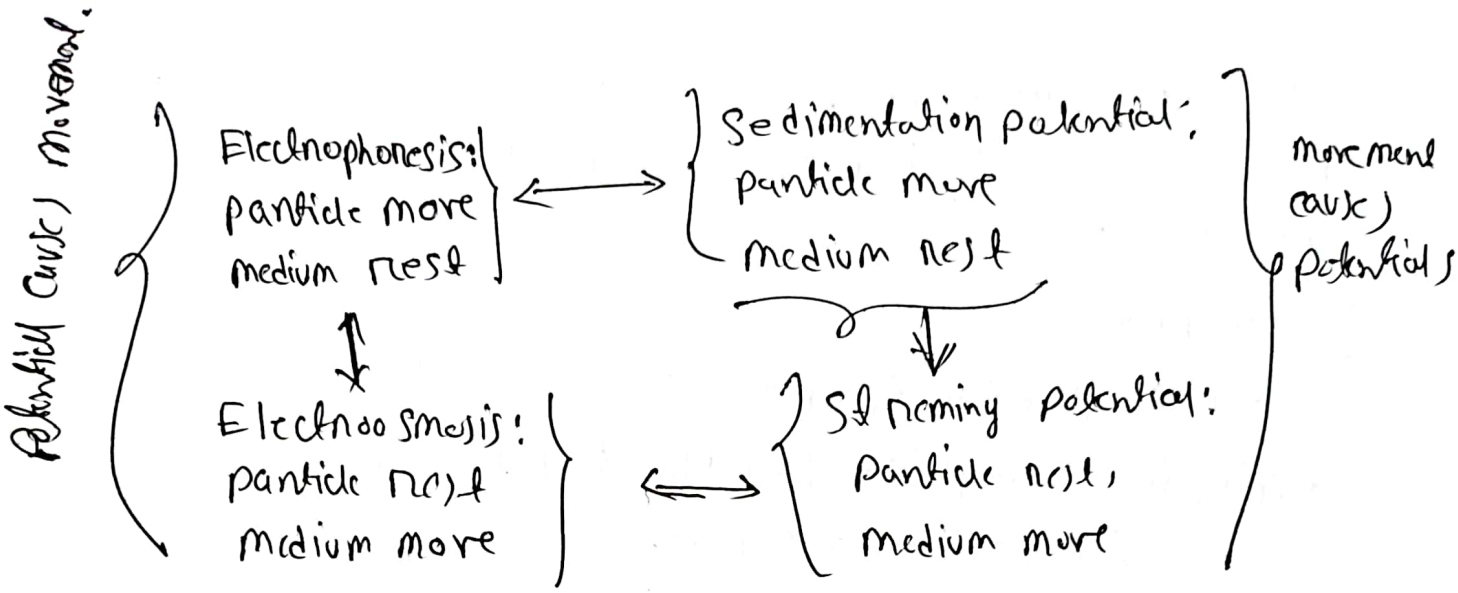


$$\epsilon = \frac{4\pi\eta v}{D}$$

ϵ = zeta potential
D = dielectric constant
η = viscosity
v = electrophonetic mobility

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(x) → Lyophilic sols may be prepared by simply warming the solid with the liquid dispersion medium. e.g Starch with water

→ on the other hand, lyophobic sols have to be prepared by special methods. These methods fall into two categories

① Dispersion method: in these methods larger macro-sized particle are broken down to colloidal size

② Aggregation methods: in these methods colloidal size particles are built up by aggregating single ions or molecules.

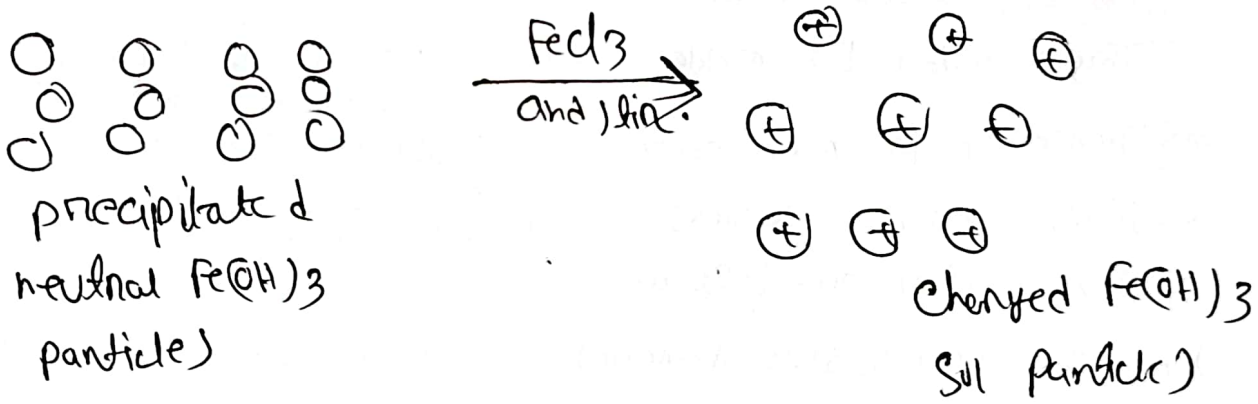
Dispersion methods:

By peptization:

→ This dispersal of a precipitated material into colloidal solution by the action of an electrolyte in solution. is termed peptization. The electrolyte used is called peptizing Agent.

→ Peptization is the reverse of **coagulation of a sol**

→ Some freshly precipitated ionic solids are dispersed into colloidal solution in water by the addition of small quantities of electrolyte, particularly those containing a common ion. The precipitate adsorbs the common ion and electrically charged particles then split from the precipitate as colloidal particles.



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